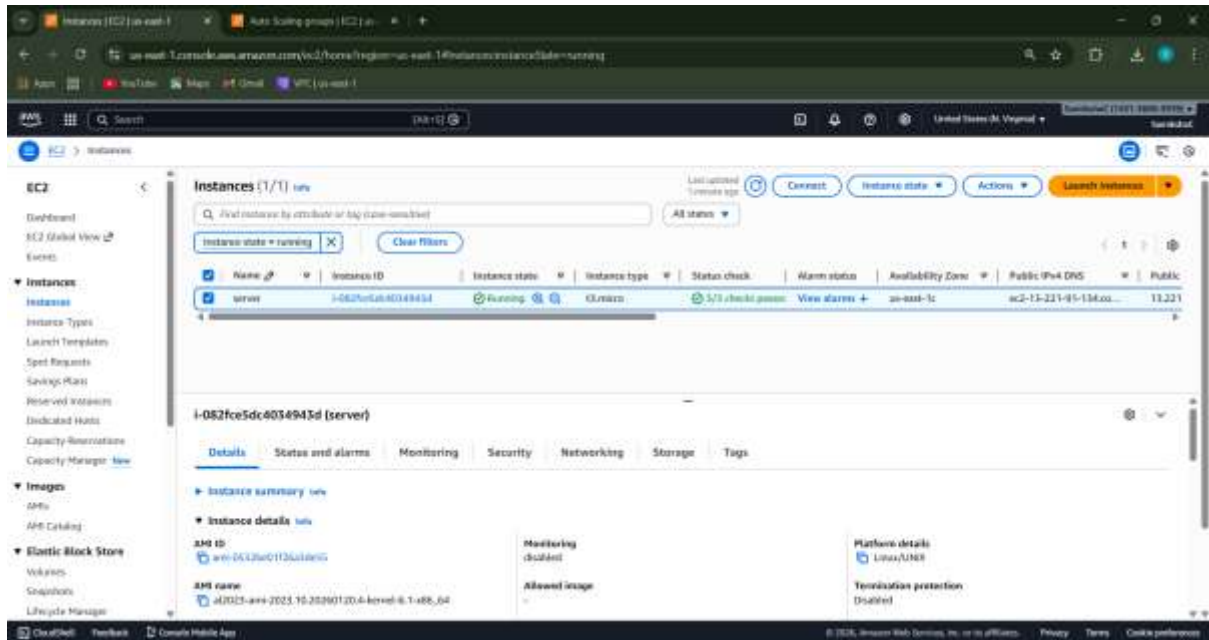


6. Cost Optimization

- Use free-tier eligible instances
- Minimal resources
- Auto Scaling to avoid over-provisioning



For more examples and ideas, visit:
<https://docs.docker.com/get-started/>

```
[root@ip-172-31-22-219 ec2-user]# mkdir ~/myapp
cd ~/myapp
[root@ip-172-31-22-219 myapp]# vi app.py
[root@ip-172-31-22-219 myapp]# vi Dockerfile
"Dockerfile" [New] 18L, 257B written
[root@ip-172-31-22-219 myapp]# docker build -t myapp:latest .
[+] Building 7.6s (9/9) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile              0.0s
=> => transferring dockerfile: 356B                             0.0s
=> [internal] load metadata for docker.io/library/python:3.11-slim 0.3s
=> [internal] load .dockerignore                                0.0s
=> => transferring context: 2B                                    0.0s
=> [1/4] FROM docker.io/library/python:3.11-slim@sha256:db27ce7778e5 3.2s
=> => resolve docker.io/library/python:3.11-slim@sha256:db27ce7778e5 0.0s
=> => sha256:8cbc47ff628d718fb76f7fca9897e4e8b607a 14.36MB / 14.36MB 0.2s
=> => sha256:d85099f0969e8b2306770a12dffcda300208cc3a18b 250B / 250B 0.1s
=> => sha256:db27ce7778e5f581d5d97812ee577a01a9fff 10.37kB / 10.37kB 0.0s
=> => sha256:543d6cace00ffc96bc95d332493bb28a4332c6d 1.75kB / 1.75kB 0.0s
=> => sha256:466c0182639be3c46be2bd95edae0a22e92ff39 5.47kB / 5.47kB 0.0s
=> => sha256:0c8d55a45c0dc58de60579b9cc5b708de9e79 29.78MB / 29.78MB 0.4s
=> => sha256:64faa99400e1388ed9f202917ada9fac34fd46c 1.29MB / 1.29MB 0.1s
=> => extracting sha256:0c8d55a45c0dc58de60579b9cc5b708de9e7957f4591 1.4s
=> => extracting sha256:64faa99400e1388ed9f202917ada9fac34fd46c958d4 0.1s
=> => extracting sha256:8cbc47ff628d718fb76f7fca9897e4e8b607a4f54300 1.0s
=> => extracting sha256:d85099f0969e8b2306770a12dffcda300208cc3a18b8 0.0s
=> [internal] load build context                                0.0s
=> => transferring context: 643B                                   0.0s
=> [2/4] WORKDIR /app                                          0.1s
=> [3/4] COPY . /app                                          0.0s
=> [4/4] RUN pip install flask                                3.5s
=> exporting to image                                          0.2s
=> => exporting layers                                           0.2s
=> => writing image sha256:43396af324ad28b0182bf590a8a6bdb09bd584fbd 0.0s
=> => naming to docker.io/library/myapp:latest                 0.0s
[root@ip-172-31-22-219 myapp]# docker run -d -p 5000:5000 myapp:latest
e6383a33d0be641475d4fdbcf1ea2d91e7bc4f5d97bf443540309929b111a94b
[root@ip-172-31-22-219 myapp]# |
```

```
ec2-user@ip-172-31-29-10:~ X root@ip-172-31-22-219:~/mye X + v
from flask import Flask
app = Flask(__name__)

@app.route('/')
def home():
    return "Hello, Cost-Optimized Docker App!"

if __name__ == '__main__':
    app.run(host='0.0.0.0', port=5000)
~
```

```
ec2-user@ip-172-31-29-10:~ X root@ip-172-31-22-219:~/mye X +
# Use lightweight Python image
FROM python:3.11-slim

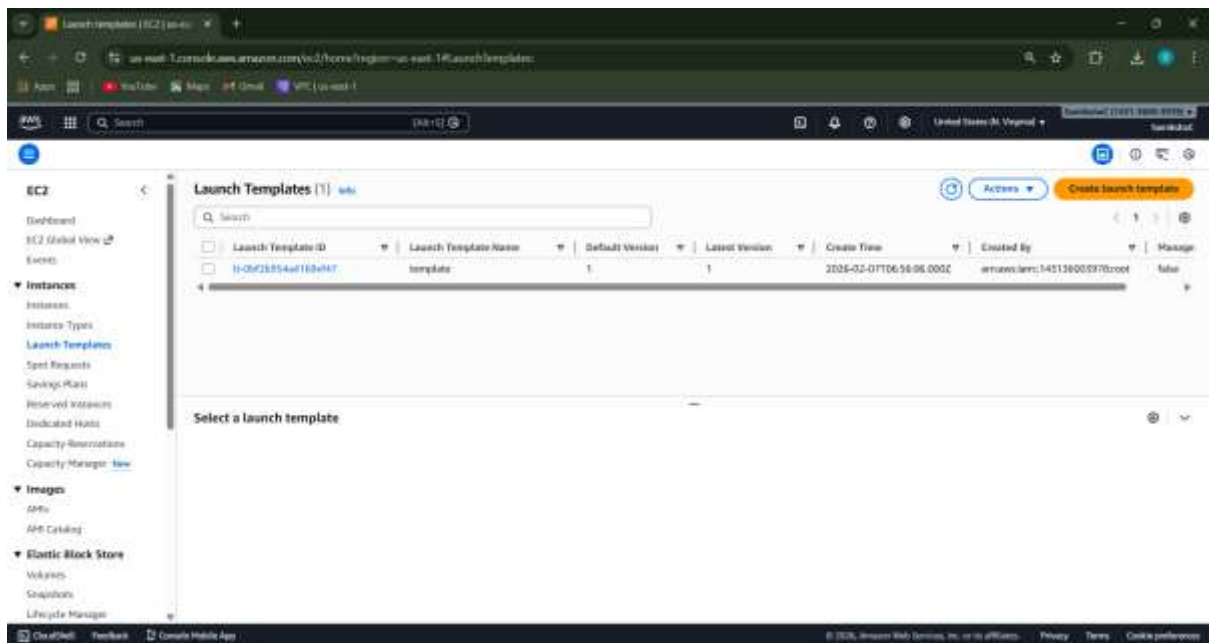
# Set working directory
WORKDIR /app

# Copy local files to container
COPY . /app

# Install dependencies
RUN pip install flask

# Expose port
EXPOSE 5000

# Command to run app
CMD ["python", "app.py"]
~
~
~
~
~
~
```



Target groups (1) [Info](#) [What's new?](#)

Filter target groups

☐	Name	ARN	Port	Protocol	Target type	Load balancer	VPC ID
☐	target	arn:aws:elasticloadbalancing:us-east-1:123456789012:targetgroup/target/12345678901234567890123456789012	80	HTTP	Instance	load-balancer	vpc-09f423702c076ac5a

0 target groups selected

Select a target group above.

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Load balancers (1) [What's new?](#)

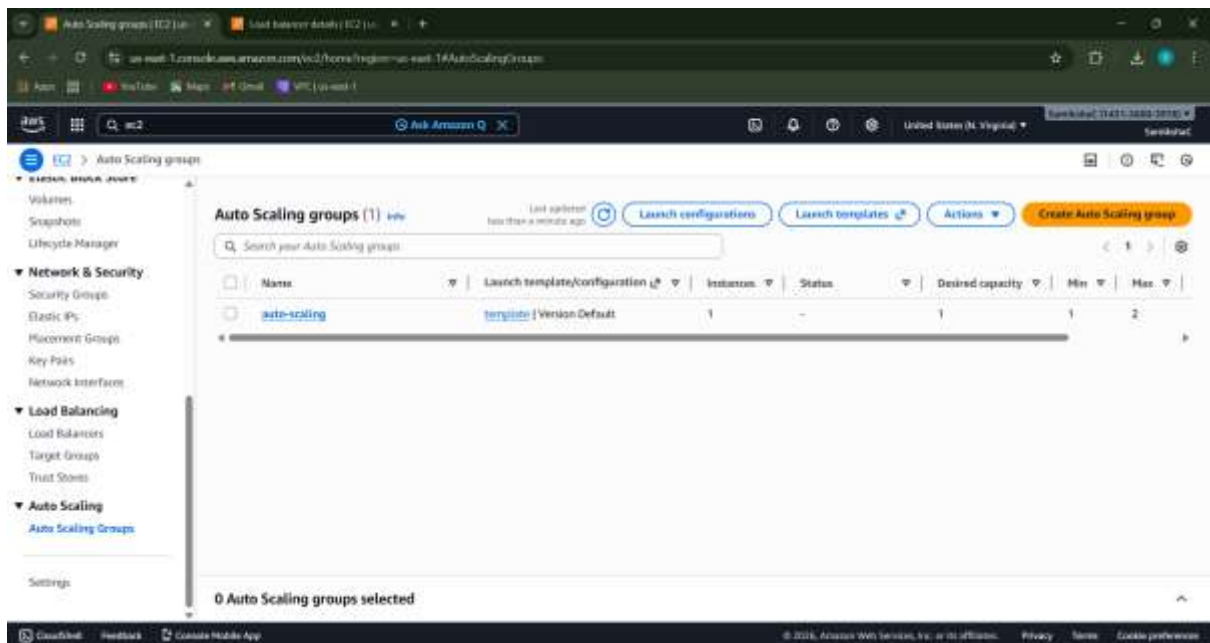
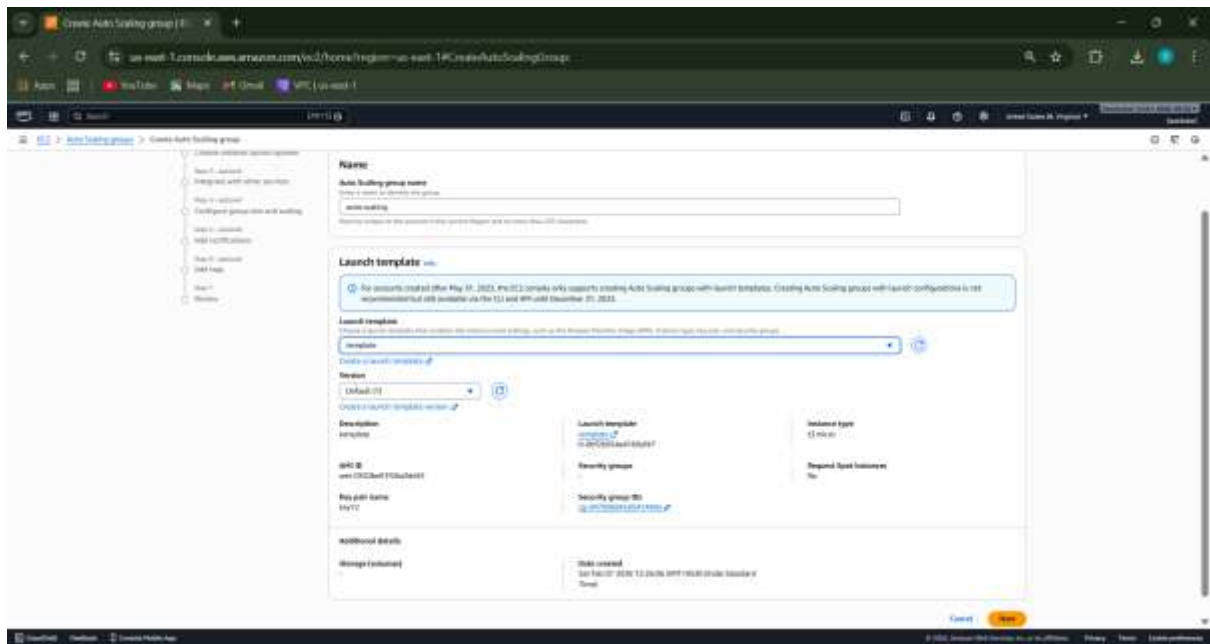
Filter load balancers

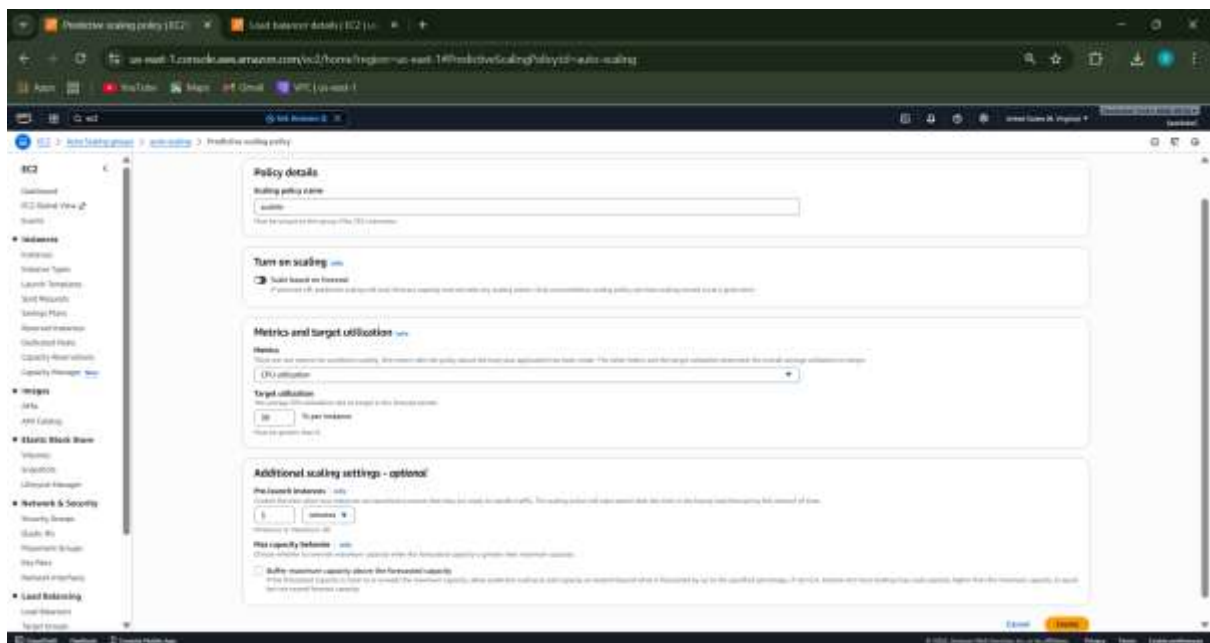
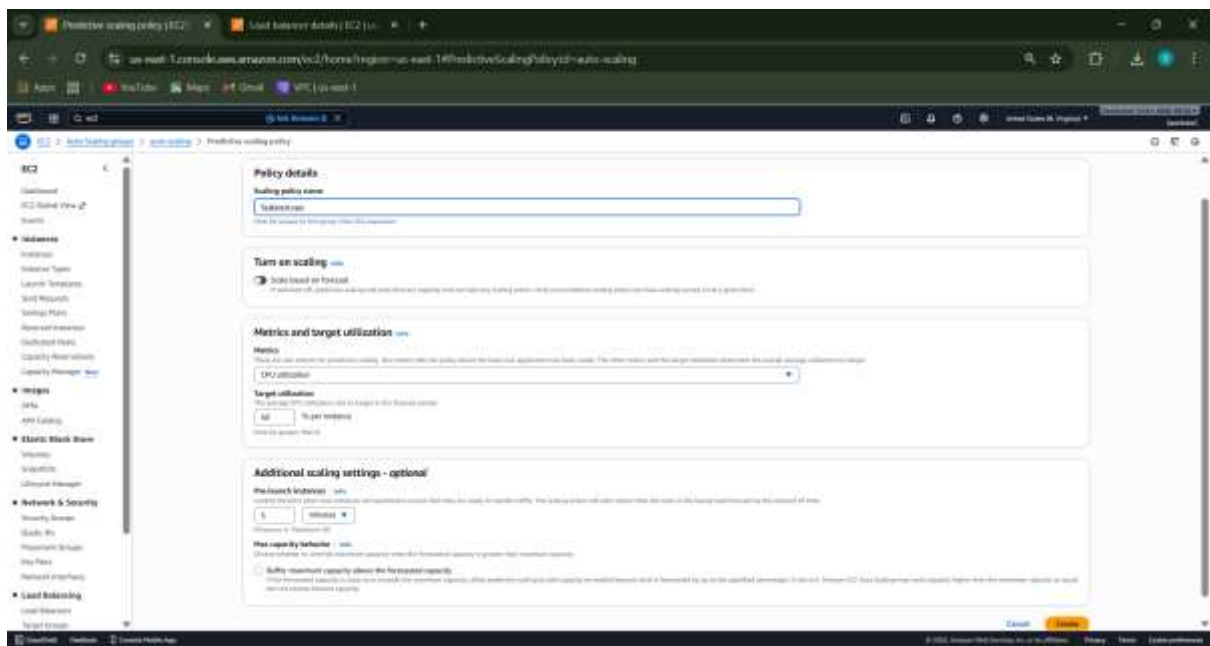
☐	Name	State	Type	Scheme	IP address type	VPC ID	Availability Zones
☐	load-balancer	Active	application	Internet-facing	IPv4	vpc-09f423702c076ac5a	5 Availability Zones

0 load balancers selected

Select a load balancer above.

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```
nc2-user@ip-172-31-28-10: ~$ sudo yum install -y stress
Last login: Sat Feb  9 06:42:43 2025 from 203.128.40.89
[ec2-user@ip-172-31-28-10 ~]$ sudo yum install -y stress
stress --cpu 1 --timeout 360
Last metadata expiration check: 0:14:26 ago on Sat Feb  9 06:43:44 2025.
Dependencies resolved.
Package Arch Version Repository Size
Installing:
stress-1.04-1.el9.aarch64 amazonlinux 36 k
Transaction Summary
Install: 1 package
Total download size: 36 k
Installed size: 88 k
Downloading Packages:
stress-1.04-1.el9.aarch64.rpm 604 KB/s | 34 kB 00:00
Total 607 KB/s | 34 kB 00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Installing : stress-1.04-1.el9.aarch64 1/1
Installing : stress-1.04-1.el9.aarch64 1/1
Verifying : stress-1.04-1.el9.aarch64 1/1
Installed:
stress-1.04-1.el9.aarch64
Complete!
stress: info: [1000] dispatching hogs: 1 cpu, 4 io, 0 vm, 0 hwm
```

